



Operating Systems

Assignment # 1

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Question No. 1. In this question, we will demonstrate the workings of some basic function calls (such as `fork()`, `wait()` and `exit()`) with a simple program while performing insertion sort.

Write a program C that takes an integer array of 10 values from the user through command line arguments. We will make a line of n processes i.e., the first process (at position 1) creates 1 child (position 2), which further creates 1 child (position 3) and so on until we have a total of 10 processes. Each process should 'receive' the array with the first (position - 1) elements sorted, use insertion sort to process the (position)th element and print the array.

No inter-process communication required. The sorting should be done inside a function, which should be registered to be called using **`atexit()`**.

Each process should wait for its child (if any). Moreover, each process should output its position in the line (the first process has position 1), its own process ID, and its parent's process ID once created and before exiting.

Question No. 2. Write a C program that inputs a number 'n', creates new file in the current directory (called `input.txt`), and enters 'n' numbers in the range 0-255 inside it. You are allowed to make small changes to the file's format for your convenience.

Afterwards, we will write a C program that

1. Calls the above program for execution.
2. Input the numbers from `input.txt` into an array and then finds the maximum number recursively using child processes. To do so, each process should calculate the size of its chunk. If it is larger than 2, it shall divide the chunk into 2 pieces (left half and right half), create 2 children (one for each chunk), and wait for them to return with the maximums in their respective chunks, the larger of which it shall pass to its parent. Otherwise, it should return the maximum in its chunk to its parent.

To return the maximums, use the `exit()` function.

Question No. 3. In this question, we explore the `exec` family of functions and touch on inter-process communication using pipes.

Study the variations in the `exec` family of functions.

(To aid your understanding:
PATH: Your OS maintains a `PATH` variable, which stores a list of directories where common executable files are stored.)

Write a C program 'double.c' that inputs a number 'n' and returns 2n. Write a C program 'square.c' that inputs a number 'n' and returns n^2 . The inputs should be taken as command line arguments.

Then, write a C program that allows the user to call the above programs in any sequence as they wish, and returns the output. For example, the user may execute `./doMath square double double 4` which would print 256.

To do so, doMath should make an unnamed pipe and make a child process which would inherit the file descriptors (make sure to close the unused ends of the pipe at this point!). Afterwards, the child process should execute (exec) the first operation ('double' in the above example). It should pass the number to operate on as well as the file descriptor of the pipe as arguments. Once it calculates the answer, it should send it to the parent (which prints it), then use exec again to execute the next operation ('double' in the above example). This process is repeated until all operations are performed.

You are allowed to make changes in 'double.c' and 'square.c' as necessary.

Guidelines:

- A single violation of guideline will lead to **Zero mark** in your assignment.
- Deadlines should be kept in mind. **No extension** in assignment dates would be given.
- Submit the `.c` or `.cpp` file and the output's screenshot in **Word** file on **Google classroom in Zip form**.
- You are encouraged to take help from Internet and books.
- You will have maximum marks if you have done the entire task.
- This is an individual assignment. **PLAGARISM IS NOT ACCEPTABLE!**
- Follow the instructions as it is, otherwise, your assignment would not be accepted at all.